



Security Auditing

Lab Session Guide Autopsy

School Year 2025/2026

Master in Informatics Security (MSI) / Master in Informatics Engineering (MEI)

Informatics Engineering Department

University of Coimbra



The Autopsy/Sleuth Kit forensic tool

Last week, we performed forensic analysis in RAM.

Today, we study how to perform forensic analysis in the filesystem.

The Autopsy forensic tool: (<https://www.sleuthkit.org/autopsy/>)

- Autopsy is a digital forensics platform and graphical interface to [The Sleuth Kit](#)
- full-featured GUI forensic suite used for analysing disk images, recovering deleted files, creating activity timelines and examining artifacts.
- It is essential for post-incident disk analysis, recovery of attacker tools, persistence mechanisms, and user activity reconstruction.

The Environment:

- The slides assume we will work inside the SIFT Workstation VM, to ensure tool compatibility for everyone.
- Nonetheless, Autopsy can be installed directly on Ubuntu and can also run on Windows (in fact, its best interface is for Windows!). Feel free to install and explore it anyway you want.



Launching Autopsy in your SIFT Workstation

Starting Autopsy in your SIFT Workstation:

- just launch Autopsy

\$sudo autopsy

- using a web browser, just open <http://localhost:9999/autopsy>
- you are now ready to explore the tool.
- go straight to slide 10 (“Testing Autopsy”) to find some interesting disk dump files and start exploring.

Please beware that some plugins are not available for ARM64 architectures, but most of the stuff should work fine.



Launching Autopsy in Your SIFT Workstation

In Windows Autopsy runs on a native GUI and is launched as any other application.

In the SIFT Workstation Autopsy runs as with a web interface on port 9999:

- Just launch auto

- https://github.com/sleuthkit/autopsy/blob/develop/linux_macos_install_scripts/install_prereqs_ubuntu.sh

- open a terminal with elevated privileges... or just copy paste this:

- [sudo curl -O https://github.com/sleuthkit/autopsy/blob/develop/linux_macos_install_scripts/install_prereqs_ubuntu.sh](https://github.com/sleuthkit/autopsy/blob/develop/linux_macos_install_scripts/install_prereqs_ubuntu.sh)

- you may also use “wget”:

We also need *PhotoRec*, that comes bundled with the *testdisk* package:

- [sudo apt update](#)

- [sudo apt install testdisk](#)



Installation and Configuration (1) Sleuth Kit

(only in case it is not ready-installed in your SIFT Workstation)

Follow the instructions available for instance at this link:

– <https://rahulcyberx.medium.com/install-use-autopsy-for-digital-forensics-on-linux-05782d5e2d26>

OR FOLLOW THE FOLLOWING INSTRUCTIONS (adapted from that link):

Installing prerequisites:

– In Ubuntu, we need to run this script designed to install necessary dependencies:

https://github.com/sleuthkit/autopsy/blob/develop/linux_macos_install_scripts/install_prereqs_ubuntu.sh

– open a terminal with elevated privileges... or just copy paste this:

`sudo curl -O https://github.com/sleuthkit/autopsy/blob/develop/linux_macos_install_scripts/install_prereqs_ubuntu.sh`

– you may also use “wget”:

```
install_prereqs_ubu [ <=> ] 185.79K --.-KB/s in 0.1s
2025-12-29 09:53:00 (1.62 MB/s) - 'install_prereqs_ubuntu.sh.1' saved [190248]
```

We also need *PhotoRec*, that comes bundled with the *testdisk* package:

`sudo apt update`

`sudo apt install testdisk`

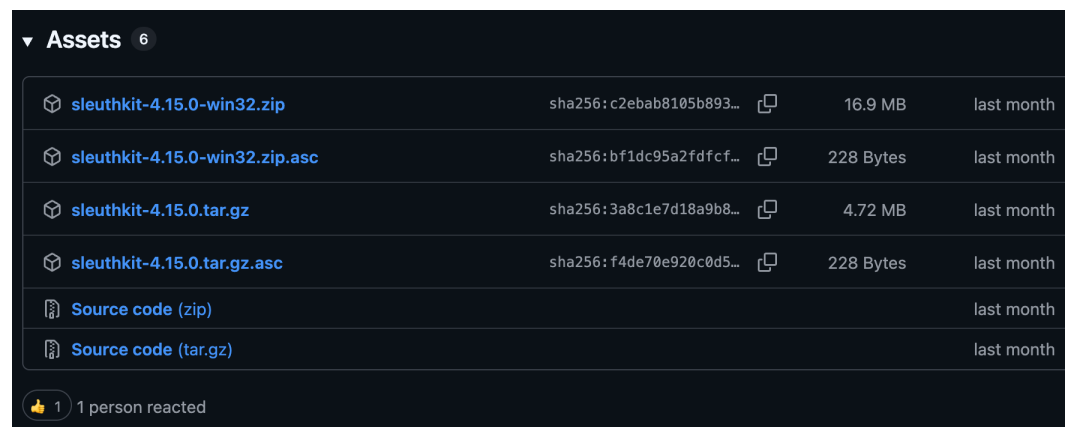


Installation and Configuration (2) Sleuth Kit

(only in case it is not ready-installed in your SIFT Workstation)

Installing the Sleuth Kit (in Ubuntu):

– In this link you will see there are various releases available: <https://github.com/sleuthkit/sleuthkit/releases>



▼ Assets 6				
 sleuthkit-4.15.0-win32.zip	sha256: c2ebab8105b893...		16.9 MB	last month
 sleuthkit-4.15.0-win32.zip.asc	sha256: bf1dc95a2fdcf...		228 Bytes	last month
 sleuthkit-4.15.0.tar.gz	sha256: 3a8c1e7d18a9b8...		4.72 MB	last month
 sleuthkit-4.15.0.tar.gz.asc	sha256: f4de70e920c0d5...		228 Bytes	last month
 Source code (zip)				last month
 Source code (tar.gz)				last month
 1 1 person reacted				

- We need to download the correct one for your computer.
- For ARM64 the link is: <https://archlinuxarm.org/packages/aarch64/sleuthkit>
- and install it: `sudo apt update && sudo apt install ./DOWNLOADED_FILE`



Installation and Configuration (3.1) Autopsy

(only in case it is not ready-installed in your SIFT Workstation)

- In order to install Autopsy, we need: Java path; autopsy ZIP file; installer script
- Before continuing, make sure you have Java installed by checking `java -version`
- If not, run this:

`sudo apt update`

`sudo apt install -y openjdk-17-jdk unzip`

- Then find Java home (needed by the installer): `update-java-alternatives -l | grep java-1.17`

– Expected result:

```
xrahul@xrahul-VirtualBox:~$ update-java-alternatives -l | grep java-1.17
java-1.17.0-openjdk-amd64      1711      /usr/lib/jvm/java-1.17.0-openjdk-amd64
xrahul@xrahul-VirtualBox:~$
```

- Copy the path.
- Now you need the Autopsy zip file from Github: <https://github.com/sleuthkit/autopsy/releases>
- Run: (adjust path and file names): `wget`
https://github.com/sleuthkit/autopsy/releases/download/your_pack.zip



Installation and Configuration (3.2) Autopsy

(only in case it is not ready-installed in your SIFT Workstation)

- Download just the script:

```
wget https://raw.githubusercontent.com/sleuthkit/autopsy/develop/linux_macos_install_scripts/install_application.sh
```

```
chmod +x install_application.sh
```

- Finally, run the installer: *(adjust paths and names)*

```
sudo ./install_application.sh \
```

```
-z ~/autopsy-4.22.1_v2.zip \
```

```
-i ~/autopsy \
```

```
-j /usr/lib/jvm/java-1.17.0-openjdk-xxxxx
```

z → path to the Autopsy zip

-i → where Autopsy will be installed

-j → Java 17 **HOME directory**, not the java binary

- If successful, you will see this:

```
Application is now configured. You can execute bin/autopsy to start it
```

```
/home/xrahul
```

```
Application setup done. You can run autopsy from /home/xrahul/autopsy/autopsy-4.22.1/bin/autopsy.
```




Running Autopsy

(only in case it is not ready-installed in your SIFT Workstation)

Option A: Run from terminal (recommended first run)

```
cd /home/xrahul/autopsy/autopsy-4.22.1/bin  
./autopsy
```

- some messages and warnings will come up in the terminal.
- it's safe to ignore them as long as it's not an error.
- after some time (20–40 seconds) you will the Autopsy splash screen, and then the GUI launches.

Option B: Add Autopsy to you default path:

```
cd ~  
echo 'export PATH=$PATH:/home/your-path/autopsy/autopsy-4.22.1/bin' >> ~/.bashrc  
source ~/.bashrc
```

- Then you can just type in terminal:

```
autopsy
```



Testing Autopsy

To test Autopsy, we need a working forensic test image:

- we will use Digital Corpora — It's small, and reliable for testing.
- create a directory which you will use to store forensic images.

mkdir -p ~/forensics

cd ~/forensics

- download the image :

wget <https://digitalcorpora.s3.amazonaws.com/corpora/drives/nps-2009-canon2/nps-2009-canon2-gen6.E01>

- now you have a very real forensic E01 image! Launch Autopsy and create a new case.

If the downloaded disk didn't work (or you want to explore other disks), use these repositories:

- Digital Corpora, <https://digitalcorpora.org/>
- NIST CFReDS Portal (Computer Forensic Reference Data Set), <https://cfreds.nist.gov/>
- DFIR Training Test Images, <https://www.dfir.training/downloads/test-images>
- Digital Forensics Workbook Data Sets, <https://www.digitalforensicsworkbook.com/data-sets>
- Digital Forensic Tool Testing (DFTT) Images, <https://dftt.sourceforge.net/>
- GitHub Sample Autopsy Case (with an .E01 image), <https://github.com/oddin-forensic/autopsy-sample-case>